

Appendix A

The notation used throughout the mathematical model is summarized in Table A1.

Table A1. Notation used in the proposed model.

Symbol:	Description:
Sets:	
H	Set of hospitals
R_h	Set of operating rooms in hospital h
D	Set of planning days
J	Set of time slots within one working day
I	Set of elective patients
I^C	Set of compulsory patients
I^O	Set of optional patients
S	Set of surgeons
S_{hd}	Set of surgeons available in hospital h on day d
s_{hp}	Set of professors in hospital h
s_{hap}	Set of associate professors in hospital h
as_{has}	Set of attending surgeons in hospital h
Indices:	
h	Hospital index
r	Operating-room index
d	Planning-day index
j	Surgery starting time slot
t	Arbitrary time slot
i	Patient index
s	Surgeon index
Parameters:	
T	Number of planning days
G_{hd}	Fixed cost of activating hospital h on day d
F_{hd}	Cost of opening one operating room in hospital h on day d
B_{hd}	Available operating time of one operating room in hospital h on day d
l_i	Surgery duration of patient i (time slots)
ρ_i	Disease urgency coefficient of patient i
α_i	Waiting days accumulated by patient i before the planning horizon
τ	Urgency threshold
δ	Expected postponement beyond the current planning horizon
Sat_{is}	Utility obtained by patient i when surgeon s performs the surgery
a_{hdst}	Availability of surgeon s in hospital h on day d during time slot t
Decision variables:	
o_{hd}	Equals 1 if hospital h is activated on day d ; 0 otherwise
y_{hdr}	Equals 1 if operating room r is opened in hospital h on day d ; 0 otherwise
w_i	

x_{ihdrs_j}	Equals 1 if optional patient i is postponed beyond the current planning horizon; 0 otherwise
	Equals 1 if patient i is scheduled in hospital h , operating room r , by surgeon s , on day d , starting at time slot j ; 0 otherwise

Appendix B

Consider a small instance involving two hospitals, $H = \{h_1, h_2\}$, over a two-day planning horizon, $D = \{1, 2\}$. Each hospital contains two identical operating rooms, denoted by $R_h = \{r_1, r_2\}$. Each working day is discretized into 32 time slots of 15 minutes. The actual operating time available to an operating room at hospital h on day d is denoted by B_{hd} . Time intervals are represented as half-open intervals; for example, $[0, 16)$ contains 16 time slots. The hospital-day parameters, including the hospital activation cost G_{hd} , operating-room opening cost F_{hd} , available operating time B_{hd} , and surgeon availability intervals, are jointly summarized in Table B1.

Table B1. Hospital, operating-room, and surgeon data for the illustrative instance.

Hospital	d	F_{hd}	G_{hd}	B_{hd}	Surgeon	time slots (d_1)	time slots (d_2)
h_1	1	1904	5968	29	s_{h_1p1}	$[0, 16)$	0
					s_{h_1ap1}	$[0, 16)$	$[16, 32)$
h_1	2	1800	4856	28	as_{h_1as1}	$[0, 32)$	$[0, 16)$
h_2	1	2142	5455	29	s_{h_2p1}	$[0, 16)$	0
					s_{h_2ap1}	$[0, 32)$	0
h_2	2	2284	5652	32	as_{h_2as1}	$[0, 16)$	$[0, 32)$

The patient set is $I = \{i_1, \dots, i_6\}$, and the corresponding surgery durations, waiting times, disease urgency coefficients, urgency scores, and baseline utilities are summarized in Table B2. The urgency scores are calculated according to the definition provided in Section 3.1. With the urgency threshold set to $\tau = 500$, patient i_2 belongs to the compulsory patient set I^C , whereas the remaining patients constitute the optional patient set I^O .

Table B2. Patient data for the illustrative instance.

Patient	l_i	α_i	ρ_i	$Score_i$	Baseline utility of professors	Baseline utility of associate professors	Baseline utility of attending surgeons
i_1	8	69	5	355	3.0	1.6	0.2
i_2	13	111	5	565	3.2	1.8	0.7
i_3	6	68	5	350	3.7	1.9	1.0
i_4	4	98	2	200	3.1	1.6	0.7
i_5	10	105	1	107	3.3	1.2	0.5
i_6	12	103	3	315	3.2	1.9	0.8

All six surgeons are assumed to be clinically qualified to perform the surgeries, such that $S_i = S$ for each patient i . Surgeons with the same professional qualification are assigned the same baseline utility for a given patient. The preference probabilities and patient utilities are calculated according to the patient utility model introduced in Section 3.2. For this illustrative instance, the preference-realization parameter is set to $\beta = 0.5$. For example, the baseline utilities of patient i_2 for professors, associate professors, and attending surgeons are 3.2, 1.8, and 0.7, respectively. Therefore, the average baseline utility of patient i_2 is $\bar{r}_{i_2} = \frac{3.2+1.8+0.7}{3} = 1.9$.

Accordingly, the relative utilities associated with professors, associate professors, and attending surgeons are $u_{i_2,shp} = \frac{3.2}{1.9}$, $u_{i_2,shap} = \frac{1.8}{1.9}$, $u_{i_2,asas} = \frac{0.7}{1.9}$.

The preference probability of patient i_2 for professor s_{h_2p1} at hospital h_2 is calculated as $\pi_{i_2,s_{h_2p1}} = \frac{\exp(3.2/1.9)}{2[\exp(\frac{3.2}{1.9}) + \exp(\frac{1.8}{1.9}) + \exp(0.7/1.9)]} = 0.2862$, where the coefficient 2 in the denominator reflects the presence of two surgeons at each professional qualification level, who have the same relative utility within each level. Since the assigned professor s_{h_2p1} has the highest preference probability among all qualified surgeons for patient i_2 , the corresponding preference-realization degree is $V_{i_2,s_{h_2p1}} = 1$. Therefore, the utility obtained by patient i_2 from the assignment of professor s_{h_2p1} is $Sat_{i_2,s_{h_2p1}} = 0.2862 \times 1 = 0.2862$.

The preference probabilities, preference-realization degrees, and patient utilities of the remaining patients are calculated analogously. Based on these utility values together with the operating-room and surgeon availability information, Table B3 presents a feasible candidate schedule for the illustrative instance.

Table B3. Patient scheduling results for the illustrative instance.

Patient	Hospital	d	OR	Surgery interval	Assigned surgeon
i_1	h_2	1	r_1	[13,21)	s_{h_2ap1}
i_2	h_2	1	r_1	[0,13)	s_{h_2p1}
i_3	h_2	1	r_1	[21,27)	s_{h_2ap1}
i_4	h_1	2	r_1	[12,16)	as_{h_1as1}
i_5	h_1	2	r_1	[16,26)	s_{h_1ap1}
i_6	h_1	2	r_1	[0,12)	as_{h_1as1}

As illustrated in Figure B1, the example jointly determines hospital activation, operating-room opening, patient scheduling, and surgeon assignment. On Day 1, hospital h_2 is activated and operating room r_1 is opened. Patients i_2 , i_1 , and i_3 are consecutively scheduled in intervals [0,13), [13,21), and [21,27), respectively, satisfying the available operating time $B_{h_21} = 29$. Patient i_2 is assigned to professor s_{h_2p1} , whereas patients i_1 and i_3 are assigned to associate professor s_{h_2ap1} . On Day 2, hospital h_1 is activated and operating room r_1 is opened. Patients i_6 , i_4 , and i_5 are scheduled in intervals [0,12), [12,16), and [16,26), respectively, satisfying $B_{h_12} = 28$. Patients i_6 and i_4 are assigned to attending surgeon as_{h_1as1} , while patient i_5 is assigned to associate professor s_{h_1ap1} . All assignments comply with the corresponding operating-room capacities and surgeon availability intervals, and no operating-room or surgeon overlap occurs. Consequently, all six patients are scheduled, the compulsory patient i_2 is covered, and the resulting objective vector is $(Z_1, Z_2, Z_3) = (14253, 1877, 0.5431)$. Figure B1 illustrates the resulting hospital activation, operating-room opening, patient scheduling, and surgeon assignment decisions.

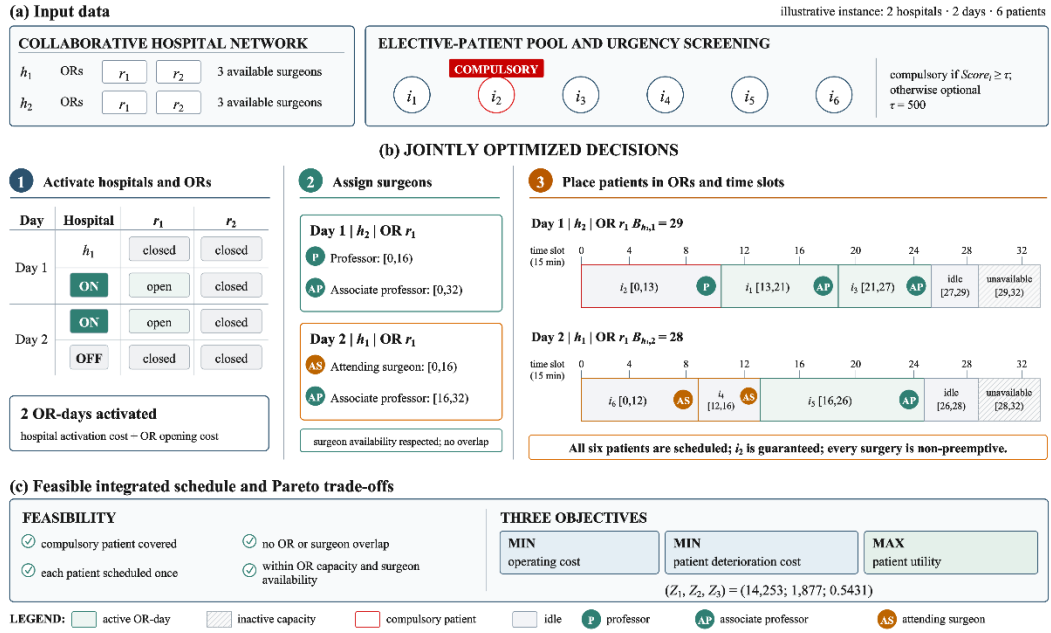


Figure B1. Operating-room schedule for the illustrative MDORSP instance.